



Computational and Differential Geometry

Homework 3

Professor: Nicolás Avilán Vargas, Ph.D.

Indicaciones

1. Fecha de entrega: 16 de mayo de 2025 hasta las 11:55 pm.
2. Único medio de entrega [e-aulas](#).
3. Formato de entrega: Un único archivo .ipynb con códigos en python, descripciones de códigos y procesos, y respuestas a las preguntas (No enviar archivos comprimidos).
4. Solo es permitido el uso de librerías “básicas” (numpy, matplotlib, plotly, seaborn, pandas, sympy, etc). En ningún caso será válida la solución lograda, total o parcialmente, por el uso de una librería especializada para resolver problemas de geometría computacional, con excepción del los diagramas de Voronoi del punto 2.
5. La **tarea** debe realizarse **individualmente**.
6. Cualquier tipo de fraude o plagio es causa de anulación directa de la evaluación y correspondiente proceso disciplinario.
7. Las entregas están sujetas a herramientas automatizadas de detección de plagio en códigos.
8. Las tareas no entregadas antes de la hora indicada tendrán calificación de 0.0.

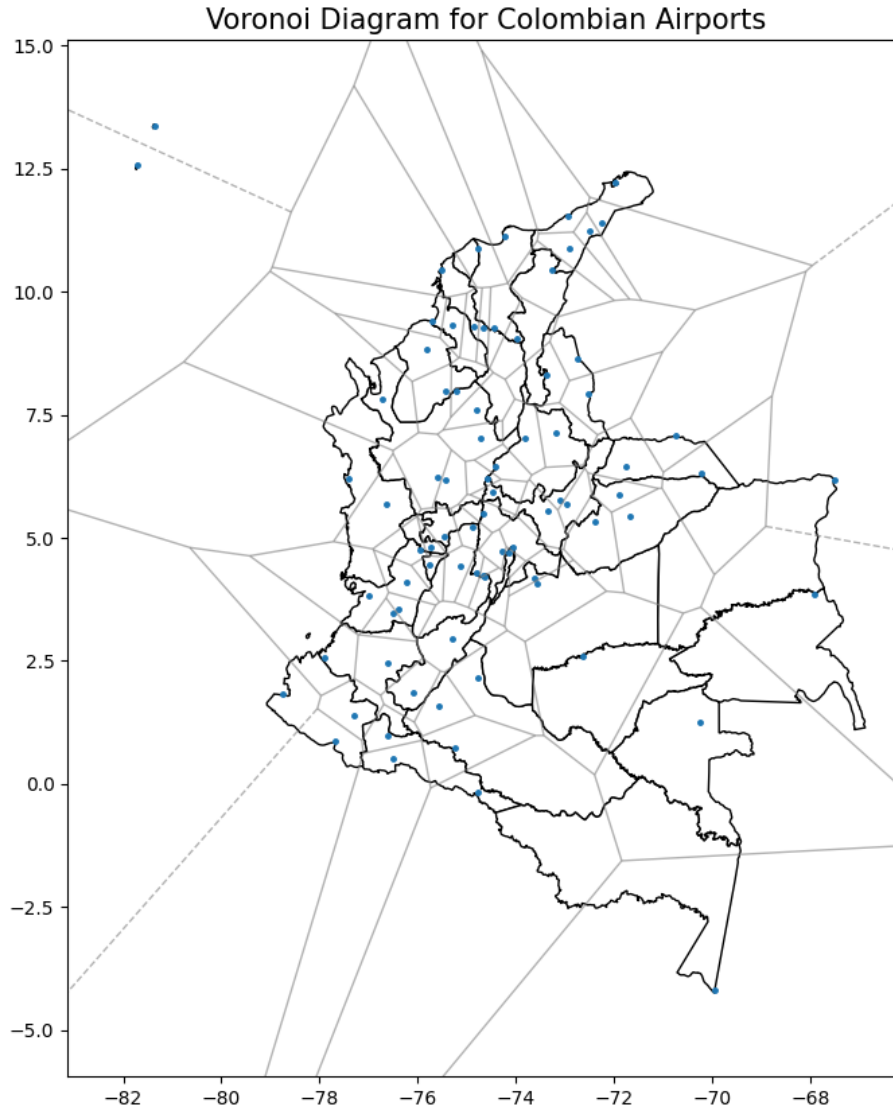
Provide a comprehensive explanation for each piece of code, covering its methods, techniques, functions, and any specific tips or tricks. Cite your sources (including papers, books, tutorials, websites, etc.), and add relevant bibliographical references or links as needed.

First part

1. (0.8 points) Occasionally, when mechanical issues arise, flights must follow an emergency protocol, including landing at the nearest airport. In such cases, knowledge of the Voronoi diagram associated with nearby airports becomes crucial, allowing the pilot to update the flight plan to identify the quickest landing option.

Read the data from the link [airports_CO.csv](#), which contains coordinates of airports in Colombia.

Use the [scipy](#) library, specifically the [Voronoi](#) and [voronoi_plot_2d](#), to plot the Voronoi diagram associated with Colombian airports on the map, as shown in the figure. You may find it helpful to explore the links: [Voronoi_KMeans](#) and [Mapas](#).



Second part

All the calculations should be developed using the Sympy library in Python.

- (0.8 points) The *evolute* of a plane curve $\alpha(t)$ is the locus of their centers of curvature and is given by

$$\beta(t) = \alpha(t) + \frac{1}{\kappa(t)} \vec{N}(t)$$

with $\vec{N}(t)$ the normal vector to the curve at t . Check out this beautiful animation: [evoluta of an ellipse](#).

Consider the curve

$$\alpha(t) = (a \cos(t), b \sin(t), 0) \quad \text{with} \quad 0 < t < 2\pi$$

- Evaluate curvature $\kappa(t)$ for $\alpha(t)$.
- Obtain the evolute for the curve $\alpha(t)$.
- Plot the curves $\alpha(t)$ and $\beta(t)$ in the same frame. Set values $a = 2$ and $b = 3$ for the plot.

3. (0.8 points) Consider the curve

$$\alpha(t) = (\cos(3t) \cos(t), \cos(3t) \sin(t), 0), \quad 0 < t < \pi$$

- Plot the curve $\alpha(t)$ with the tangent vector when $t = 0$.
- Plot the curvature $\kappa(t)$.
- Evaluate the curve's length.
- Evaluate the rotation index.

$$I = \frac{1}{2\pi} \int_C \kappa(t) \|\alpha'(t)\| dt$$

Use numerical integration if necessary. Interpret your result.

4. (0.8 points) Consider the curve

$$\alpha(t) = \left(\frac{\cos(t)}{1 + \sin^2(t)}, \frac{\sin(t) \cos(t)}{1 + \sin^2(t)}, 0 \right), \quad 0 < t < 2\pi$$

- Plot the curve $\alpha(t)$ with the tangent vector when $t = 0$.
- Plot the curvature $\kappa(t)$.
- Evaluate the curve's length.
- Evaluate the rotation index. Use numerical integration if necessary. Interpret your result.

5. (0.8 points) Show that torsion $\tau(t)$ for a non-arc length parametrized curve $\alpha(t)$ is given by

$$\tau(t) = -\frac{(\alpha'(t) \times \alpha''(t)) \cdot \alpha'''(t)}{|\alpha'(t) \times \alpha''(t)|^2}$$

Include the procedure in your notebook using L^AT_EX.

6. (1 points) Consider the toroidal spiral curve

$$\alpha(t) = ((a + \cos(\omega t))\cos(t), (a + \cos(\omega t))\sin(t), \sin(\omega t))$$

- Evaluate the unit vectors $\vec{T}(t)$, $\vec{N}(t)$ and $\vec{B}(t)$ for $\alpha(t)$.
- Evaluate and plot curvature $\kappa(t)$ and torsion $\tau(t)$ for $\alpha(t)$ when parameters are $a = 4$ and $\omega = 10$.
- Plot the curve $\alpha(t)$ and add the vectors $\vec{T}(0)$, $\vec{N}(0)$ and $\vec{B}(0)$.

Submit: Upload to the platform an **.ipynb** file with answers, codes, descriptions and plots.